INCH-POUND

MIL-C-11272/1E
4 February 2008
SUPERSEDING
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7 February 2003

MILITARY SPECIFICATION SHEET

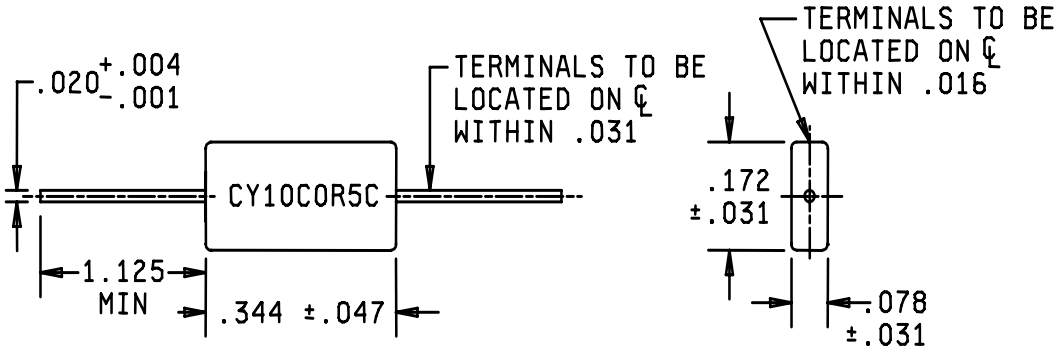
CAPACITORS, FIXED, GLASS DIELECTRIC
STYLE CY10

This specification is approved for use by all Departments and Agencies of the Department of Defense.

* The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-C-11272.

*

INACTIVE FOR DESIGN AFTER
7 JAN 1970. USE MIL-PRF-23269/1.




| Inches | mm |
|---|---|
| .001 | 0.03 |
| .004 | 0.10 |
| .016 | 0.41 |
| .020 | 0.51 |
| .031 | 0.79 |
| .047 | 1.19 |
| .078 | 1.98 |
| .172 | 4.37 |
| .344 | 8.74 |
| 1.125 | 28.58 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 1. Style CY10 capacitors.



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REQUIREMENTS:

Requirements: Requirements shall be in accordance with MIL-C-11272, and as specified herein.

* Dimensions and configuration: See figure 1.

Capacitance: See table I.

TABLE I. Style CY10 Electrical Characteristics.

| PIN 1/ | Capacitance | DC rated voltage | Capacitance tolerances available | PIN 1/ | Capacitance | DC rated voltage | Capacitance tolerances available |
|---|---|---|---|---|---|---|---|
| | pF | volts | | | pF | volts | |
| CY10C0R5C | 0.5 | 500 | C | CY10C240- | 24 | 500 | GJ |
| CY10C1R0C | 1.0 | 500 | C | CY10C270- | 27 | 500 | FGJ |
| CY10C1R5C | 1.5 | 500 | C | CY10C300- | 30 | 500 | FGJ |
| CY10C2R2- | 2.2 | 500 | CD | CY10C330- | 33 | 500 | FGJ |
| CY10C2R7C | 2.7 | 500 | C | CY10C360- | 36 | 500 | FGJ |
| CY10C3R0- | 3.0 | 500 | CD | CY10C390- | 39 | 500 | FGJ |
| CY10C3R3C | 3.3 | 500 | C | CY10C430- | 43 | 500 | FGJ |
| CY10C3R6- | 3.6 | 500 | CD | CY10C470- | 47 | 500 | FGJ |
| CY10C3R9C | 3.9 | 500 | C | CY10C510- | 51 | 500 | FGJ |
| CY10C4R3- | 4.3 | 500 | CD | CY10C560- | 56 | 500 | FGJ |
| CY10C4R7C | 4.7 | 500 | C | CY10C620- | 62 | 500 | FGJ |
| CY10C5R1C | 5.1 | 500 | C | CY10C680- | 68 | 500 | FGJ |
| CY10C5R6- | 5.6 | 500 | CJ | CY10C750- | 75 | 500 | FGJ |
| CY10C6R2- | 6.2 | 500 | CJ | CY10C820- | 82 | 500 | FGJ |
| CY10C6R8- | 6.8 | 500 | CJ | CY10C910- | 91 | 500 | FGJ |
| CY10C7R5- | 7.5 | 500 | CJ | CY10C101- | 100 | 500 | FGJ |
| CY10C8R2- | 8.2 | 500 | CJ | CY10C111- | 110 | 500 | FGJ |
| CY10C9R1- | 9.1 | 500 | CJ | CY10C121- | 120 | 500 | FGJ |
| CY10C100- | 10 | 500 | CJ | CY10C131- | 130 | 500 | FGJ |
| CY10C110- | 11 | 500 | CJ | CY10C151- | 150 | 500 | FGJ |
| CY10C120- | 12 | 500 | CJ | CY10C161- | 160 | 500 | FGJ |
| CY10C130- | 13 | 500 | GJ | CY10C181- | 180 | 500 | FGJ |
| CY10C150- | 15 | 500 | GJ | CY10C201- | 200 | 500 | FGJ |
| CY10C160- | 16 | 500 | GJ | CY10C221- | 220 | 300 | FGJ |
| CY10C180- | 18 | 500 | GJ | CY10C241- | 240 | 300 | FGJ |
| CY10C200- | 20 | 500 | GJ | CY10C271- | 270 | 300 | FGJ |
| CY10C220- | 22 | 500 | GJ | CY10C301- | 300 | 300 | FGJ |

1/ Complete type designation will include an additional letter symbol to indicate capacitance tolerance, where applicable.



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Capacitance tolerance: C = ± 0.25 pF; D = ± 0.50 pF; F = ±1%; G = ±2%; J = ±5%.

* DC rated voltage: 300 and 500 volts (see Table I).

Temperature range: -55°C to +125°C.

* Life: In accordance with MIL-C-11272 and the following:

* For qualification and group B continuation tests:

At 25°C:

Insulation resistance: 100,000 megohms minimum.

Capacitance: Change not more than 0.5 percent of nominal value or 0.5 pF whichever is greater, from initial 25°C value. *

Dissipation factor: Not to exceed 0.0025.

At 125°C:

Insulation resistance: 10,000 megohms minimum

Capacitance: Change not more than 0.5 percent of nominal value or 0.5 pF whichever is greater, from initial 125°C value.

Dissipation factor: Not to exceed 0.007.

For performance check:

Insulation resistance: 100,000 megohms minimum.

Capacitance: Change not more than 0.5 percent of nominal value or 0.5 pF whichever is greater, from initial 25°C value. *

Dissipation factor: Not to exceed 0.0025.

Temperature coefficient and capacitance drift: The temperature coefficient and capacitance drift shall be within the limits specified in table II.

TABLE II. Temperature coefficient and capacitance drift.

| Temperature coefficient | Capacitance drift (-55° to +125°C) |
|---|---|
| parts/million/°C<br>140 ±25 | 0.1 percent or 0.1 pF, whichever is greater. |

* Referenced documents. In addition to MIL-C-11272, this document also references the following:

MIL-PRF-23269/1



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* Changes from previous issue. The margins of this specification sheet are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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